

The Discrete Advection Operator

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Aditya G. Nair · University of Nevada, Reno

1 Role in the Fractional-Step Scheme

The advection operator $\mathbf{A}$ appears in Stage 1 of the fractional-step method:

$$\mathbf{R} \mathbf{u}^F = \mathbf{S} \mathbf{u}^n + \frac{\Delta t}{2} (3\mathbf{A}(\mathbf{u}^n) - \mathbf{A}(\mathbf{u}^{n-1})) + \Delta t \nu \mathbf{b} \mathbf{c}_L, \quad (1)$$

where the $\frac{3}{2}\mathbf{A}^n - \frac{1}{2}\mathbf{A}^{n-1}$ weighting is Adams-Bashforth 2 (AB2) for the explicit advective term, and $\mathbf{R} = \mathbf{I} - \frac{\Delta t \nu}{2} \mathbf{L}$, $\mathbf{S} = \mathbf{I} + \frac{\Delta t \nu}{2} \mathbf{L}$.

Key Idea

Sign convention. We define $\mathbf{A}$ as the *positive* divergence: $\mathbf{A}(\mathbf{u}) = \nabla \cdot (\mathbf{u} \otimes \mathbf{u})$, so the momentum equation is $\partial_t \mathbf{u} + \mathbf{A}(\mathbf{u}) = -\nabla p + \nu \nabla^2 \mathbf{u}$.

2 Two Forms: Convective vs. Conservative

For incompressible flow ($\nabla \cdot \mathbf{u} = 0$), both forms are analytically identical:

$$\text{Convective: } (\mathbf{u} \cdot \nabla) \mathbf{u} \quad (\text{non-conservative}) \quad (2)$$

$$\text{Conservative: } \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) \quad (\text{divergence form}) \quad (3)$$

Expanding A_x to see why they are equal:

$$\underbrace{\frac{\partial u^2}{\partial x} + \frac{\partial(uv)}{\partial y}}_{\text{conservative}} = \underbrace{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y}}_{\text{convective}} + \underbrace{u \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)}_{=0} \quad (4)$$

Warning

The two forms are analytically equivalent but **not discretely equivalent**. Only the conservative (divergence) form:

1. Conserves momentum discretely (telescoping sums).
2. Satisfies the energy identity $\mathbf{u}^\top \mathbf{A}(\mathbf{u}) \approx 0$ for divergence-free $\mathbf{u}$.
3. Uses consistent $\pm \frac{1}{2}$ stencils on the staggered grid.

Always use the conservative form.

3 Staggered Grid and Index Convention

Index ranges (interior DOF only): u -nodes: $i = 2 \dots n_x$, $j = 1 \dots n_y$; v -nodes: $i = 1 \dots n_x$, $j = 2 \dots n_y$.

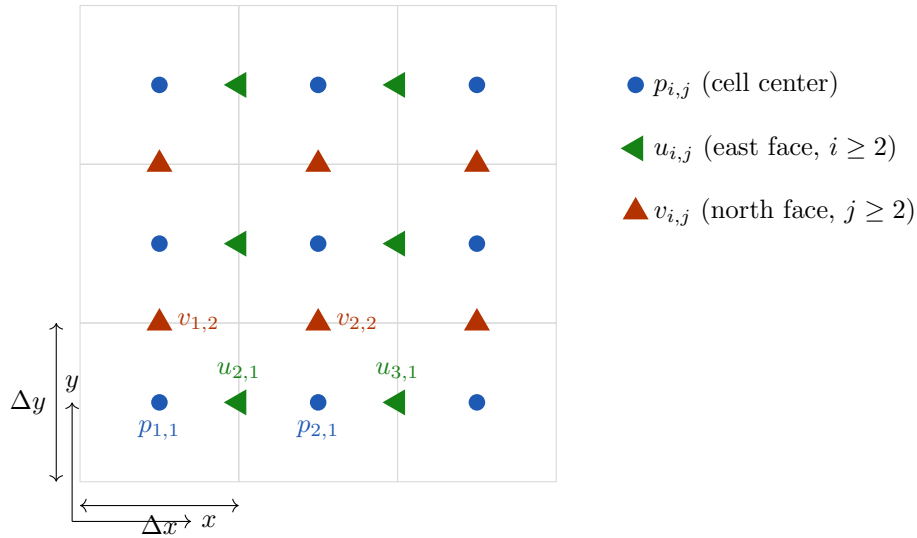


Figure 1: 3×3 staggered grid. $u_{i,j}$ enters cell (i,j) from the left; $v_{i,j}$ enters from below.

4 Discretization: Three Steps

4.1 Step 1 – Identify the control volume

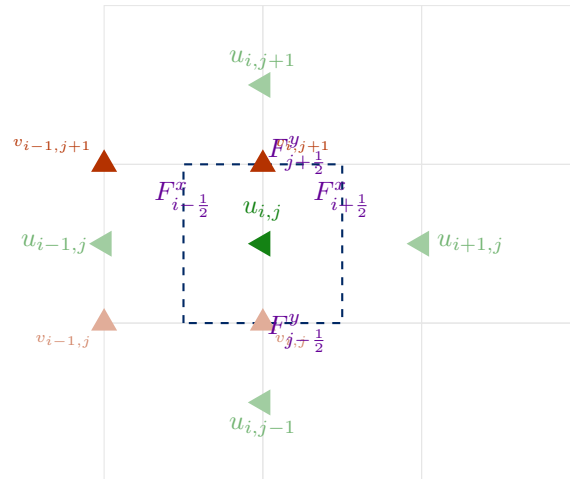


Figure 2: Control volume (dashed) for $u_{i,j}$. F^x fluxes carry u^2 ; F^y fluxes carry uv .

4.2 Step 2 – Interpolate velocities to faces ($\pm \frac{1}{2}$ rule)

Key Idea

Average the two nodes adjacent to a face *in the direction normal to that face*. This is second-order and uses a stencil of width $\pm \frac{1}{2}$ – the only consistent choice on a staggered grid.

U-component – face averages at node (i,j) :

Face	Symbol	Interpolation	Nodes used
Right x -face	$\bar{u}_{i+\frac{1}{2},j}^x$	$(u_{i,j} + u_{i+1,j})/2$	u-neighbors in x
Left x -face	$\bar{u}_{i-\frac{1}{2},j}^x$	$(u_{i-1,j} + u_{i,j})/2$	u-neighbors in x
Top y -face (u)	$\bar{u}_{i,j+\frac{1}{2}}^y$	$(u_{i,j} + u_{i,j+1})/2$	u-neighbors in y
Top y -face (v)	$\bar{v}_{i,j+\frac{1}{2}}^x$	$(v_{i-1,j+1} + v_{i,j+1})/2$	v-nodes at row $j+1$
Bot y -face (u)	$\bar{u}_{i,j-\frac{1}{2}}^y$	$(u_{i,j-1} + u_{i,j})/2$	u-neighbors in y
Bot y -face (v)	$\bar{v}_{i,j-\frac{1}{2}}^x$	$(v_{i-1,j} + v_{i,j})/2$	v-nodes at row j

Why two velocities on each y -face? The uv flux requires both a u -value (y-direction average of u) and a v -value (x-direction average of v). The v-nodes at the top face of u-cell (i, j) are $v_{i-1,j+1}$ and $v_{i,j+1}$ – they straddle the u-node in x and are always valid for $i \geq 2$ since $i - 1 \geq 1$.

4.3 Step 3 – Compute fluxes and take the divergence

The four fluxes are:

$$\begin{aligned}
 F_{i+\frac{1}{2}}^x &= (\bar{u}_{i+\frac{1}{2},j}^x)^2 & F_{i-\frac{1}{2}}^x &= (\bar{u}_{i-\frac{1}{2},j}^x)^2 \\
 F_{j+\frac{1}{2}}^y &= \bar{u}_{i,j+\frac{1}{2}}^y \cdot \bar{v}_{i,j+\frac{1}{2}}^x & F_{j-\frac{1}{2}}^y &= \bar{u}_{i,j-\frac{1}{2}}^y \cdot \bar{v}_{i,j-\frac{1}{2}}^x
 \end{aligned}$$

The discrete divergence (the final result):

$$A_x|_{i,j} = \frac{F_{i+\frac{1}{2}}^x - F_{i-\frac{1}{2}}^x}{\Delta x} + \frac{F_{j+\frac{1}{2}}^y - F_{j-\frac{1}{2}}^y}{\Delta y} \quad (5)$$

This is a **telescoping sum**: the flux leaving face $(i + \frac{1}{2})$ of cell (i, j) is the same object as the flux entering face $(i - \frac{1}{2})$ of cell $(i + 1, j)$, ensuring momentum conservation.

5 V-Component

For the v -component at node (i, j) , the same three steps apply. The key geometry difference is:

Face	Symbol	Interpolation	Notes
Right x -face (v)	$\bar{v}_{i+\frac{1}{2},j}^x$	$(v_{i,j} + v_{i+1,j})/2$	
Right x -face (u)	$\bar{u}_{i+\frac{1}{2},j}^y$	$(u_{i+1,j} + u_{i+1,j-1})/2$	rows j and $j-1$
Left x -face (v)	$\bar{v}_{i-\frac{1}{2},j}^x$	$(v_{i-1,j} + v_{i,j})/2$	
Left x -face (u)	$\bar{u}_{i-\frac{1}{2},j}^y$	$(u_{i,j} + u_{i,j-1})/2$	rows j and $j-1$
Top y -face	$\bar{v}_{i,j+\frac{1}{2}}^y$	$(v_{i,j} + v_{i,j+1})/2$	
Bottom y -face	$\bar{v}_{i,j-\frac{1}{2}}^y$	$(v_{i,j-1} + v_{i,j})/2$	

Critical geometry. The v-node (i, j) sits between rows $j-1$ and j . The u-nodes straddling the right vertical face are $u_{i+1,j}$ (above) and $u_{i+1,j-1}$ (below). This index pattern j and $j-1$ is different from the u-component case and is the most common source of indexing bugs.

$$A_y|_{i,j} = \frac{\bar{u}_{i+\frac{1}{2}}^y \cdot \bar{v}_{i+\frac{1}{2}}^x - \bar{u}_{i-\frac{1}{2}}^y \cdot \bar{v}_{i-\frac{1}{2}}^x}{\Delta x} + \frac{(\bar{v}_{j+\frac{1}{2}}^y)^2 - (\bar{v}_{j-\frac{1}{2}}^y)^2}{\Delta y} \quad (6)$$

6 Boundary Condition Treatment

When a stencil face lies on a domain wall, the missing neighbor velocity is a boundary condition. It is embedded directly into the face average:

$$\bar{u}_{\text{wall face}}^x = \frac{u_{\text{BC}} + u_{\text{interior}}}{2} \quad (7)$$

Key Idea

No ghost nodes, no separate BC vector. Unlike the Laplacian operator (which uses a separate $\mathbf{bc}_L$ vector), the advection operator embeds BCs directly because the flux is nonlinear: the boundary velocity participates as a multiplicative factor. This is precomputed each time-step since BCs may vary in time.

All wall cases, u-component:

Condition	Missing node	Replacement	Code line
$i = n_x$ (right wall)	$u_{i+1,j}$	$\mathbf{uBC_R}(j)$	$\mathbf{u_R} = (\mathbf{qi}(\mathbf{i}\mathbf{u}(i,j)) + \mathbf{uBC_R}(j))/2$
$i = 2$ (left wall)	$u_{i-1,j}$	$\mathbf{uBC_L}(j)$	$\mathbf{u_L} = (\mathbf{uBC_L}(j) + \mathbf{qi}(\mathbf{i}\mathbf{u}(i,j)))/2$
$j = n_y$ (top wall)	$u_{i,j+1}$	$\mathbf{uBC_T}(i)$	$\mathbf{u_y_T} = (\mathbf{qi}(\mathbf{i}\mathbf{u}(i,j)) + \mathbf{uBC_T}(i))/2$
$j = n_y$ (top wall)	$v_{i,j+1}$	$\mathbf{vBC_T}$	$\mathbf{v_x_T} = (\mathbf{vBC_T}(i-1) + \mathbf{vBC_T}(i))/2$
$j = 1$ (bot wall)	$u_{i,j-1}$	$\mathbf{uBC_B}(i)$	$\mathbf{u_y_B} = (\mathbf{uBC_B}(i) + \mathbf{qi}(\mathbf{i}\mathbf{u}(i,j)))/2$
$j = 1$ (bot wall)	$v_{i,j}$	$\mathbf{vBC_B}$	$\mathbf{v_x_B} = (\mathbf{vBC_B}(i-1) + \mathbf{vBC_B}(i))/2$

7 The Wrong Way: Why Not Use the ± 1 Stencil?

It is tempting to difference across a full cell instead of a half cell:

$$A_x^{\text{wrong}}|_{i,j} \approx \frac{u_{i+1,j}^2 - u_{i-1,j}^2}{2\Delta x} + \frac{u_{i,j+1} \cdot \bar{v}_{i,j+1}^{xy} - u_{i,j-1} \cdot \bar{v}_{i,j-1}^{xy}}{2\Delta y}$$

Warning

This ± 1 stencil is wrong on the staggered grid because:

1. The x -difference is second-order, but the y -difference requires a double interpolation of v (average in x then in y), which is only first-order for general fields.
2. The two directions have different effective accuracy, breaking directional symmetry.
3. Interior fluxes no longer cancel in pairs, so momentum is not conserved.

This is Slide 49 in the lecture notes – labelled “the incorrect way.”

8 Properties of the Discrete Operator

8.1 Momentum conservation

Summing Eq. (5) over all u-nodes, interior fluxes telescope:

$$\sum_{i,j} A_x|_{i,j} \Delta x \Delta y = \sum_{\text{boundary faces}} F^x \cdot \hat{n} \Delta y$$

This is the discrete divergence theorem. No momentum is created or destroyed in the interior.

8.2 Energy consistency

For a discretely divergence-free field, the conservative form satisfies:

$$\mathbf{u}^\top \mathbf{A}(\mathbf{u}) \approx 0 \quad (O(\Delta x^2) \text{ error from discrete div-free residual})$$

A non-conservative discretization gives an $O(1)$ error here, acting as a hidden source of kinetic energy.

8.3 Second-order accuracy

Each face average is second-order: $\bar{u}_{i+\frac{1}{2}}^x = u(x_{i+\frac{1}{2}}) + O(\Delta x^2)$. Combined with the second-order central difference for the divergence, the operator is globally $O(\Delta x^2)$.